

Prelude to the Process Chapters

Highlights

- How UX processes, methods, and techniques intertwine over time
- The UX design studio
- Project commission: How does a project start?
- Key UX roles
- The Middleburg University Ticket Transaction Service and the new Ticket Kiosk System: Our running examples for the process chapters
- How to write a clear and concise product concept statement

5.1 INTERTWINING OF PROCESSES, METHODS, AND TECHNIQUES

Although we present the UX lifecycle activities in separate chapters to focus on learning about each one at a time, the reality of UX practice is that the process is a fabric of interwoven lifecycle activities, methods, and techniques, not performed in a nice orderly sequence but intertwined and overlapped within the process.

5.1.1 Examples of Intertwining

Consider prototyping, which is featured as a major UX design lifecycle activity in a chapter of its own. Prototypes, however, which are tangible manifestations of any ideas that need to be evaluated, appear in many different guises in varied places.

Multiple low-fidelity rapid prototypes called sketches are used to explore competing design ideas in ideation (a kind of brainstorming; see [Sections 2.3.4.1](#) and [13.3.6](#)). Later, a more complete and detailed prototype will be used as the platform for UX evaluation of a product or system (see [Part 5](#)).

As a concrete example, a wireframe, a skeletal representation of the layout of interaction objects (e.g., menus, buttons; see [Section 18.3](#)), is an important kind of prototype. Quick and dirty wireframes (templates of screens or webpage designs showing some layout and navigation) are used as sketches for exploring design ideas as part of early design. Finished and detailed wireframes are used to convey design ideas to stakeholders and (in [Chapters 16 and 18](#)) as specifications to software implementers.

5.1.2 Lifecycle Activity Timing

Another area where practice strays from a pure lifecycle activity description is the timing of how things occur. For example, in our UX research data gathering discussion (observation and interviewing of users; see [Chapter 6](#)), it might seem that all user visits happen in one contiguous time period; however, in practice, that is rarely the case. The data elicitation activity can occur on and off over a period of time because of a variety of project constraints, including scheduling, user availability, and concurrent and unrelated things happening simultaneously. In our experience, it never seemed feasible to do all data elicitation in one go, even if the team had enough UX researchers. Instead, it was more natural to gather data with a few users and start early data analysis, modeling, synthesis, and even some design while waiting for the next user visit. Whenever convenient, the team would update the models and the design ideas with any new UX research data.

Similarly, sometimes when we are still doing data elicitation, we want to try a few early designs and prototypes that we would take to users to confirm we are on the right track. Here is a typical, more agile sequence of activities we have seen in practice:

- Data elicitation with a few early users
- Analysis and modeling of data collected
- Ideation and sketching for generative design based on analysis and emerging models
- Early wireframes or prototypes of emerging designs
- Data elicitation from more users:
 - Show users emerging design ideas and wireframes at the end of elicitation sessions for feedback
- Update user data models and designs with new insights
- Increase fidelity of wireframe details as confidence on emerging designs grows
- Evaluate with more users using more rigorous methods (e.g., in a UX lab)
- Update user data models and designs with evaluation findings
- Design production and hand off to the software engineering (SE) team

For clarity of exposition in this book, however, we break out each UX design activity and isolate it in its own chapter. This facilitates immersion by focusing the reader on a single topic at a time and helps readers use this book as a reference while doing a particular UX activity.

5.2 A DEDICATED UX DESIGN STUDIO, AN ESSENTIAL TOOL FOR TEAMWORK

Before we begin exploring the UX process chapters, we take this time to talk about an indispensable tool for every UX group: the UX design studio.

5.2.1 Why You Need a UX Design Studio

As Bødker and Buur (2002) urge, every UX group needs to set aside a shared physical workspace, a design studio, as a home base for the team. A UX design studio provides space for:

- Colocation to support constant collaboration (Brown, 2008)
- Getting deeply immersed in the design problem and the artifacts of analysis and design
- Ideation for design
- Permanent display of work products for discussion and brainstorming

Of these, perhaps the most important point is about immersion. When you walk in the room and close the door, you need to be in the world of the design, while the rest of the world and its distractions disappear.

5.2.2 What You Need in Your UX Design Studio

A good design studio will have:

- A door that can be closed
- Comfortable seating
- A conference table
- Worktables to lay out prototypes and other design artifacts
- Whiteboards for group sketching
- Ample wall space for hanging posters, drawings, diagrams, and design sketches
- Large shared display devices on which team members can show work artifacts from their laptops
- Dedicated workspace that:
 - Is constantly available without the need for reservations
 - Allows all artifacts to remain displayed from one work session to another

- Meets the unpredictable need for frequent and immediate short spurts of collaboration

5.2.3 The Virginia Tech Industrial Design Studio: The Kiva

In Fig. 5.1, we show an example of a collaborative ideation and design studio, called the Kiva, in the Virginia Tech Department of Industrial Design. The Kiva was originally designed and developed by Maya Design in Pittsburgh and is used at Virginia Tech with their permission.

The Kiva is a cylindrical space in which designers can brainstorm and sketch in isolation from outside distractions. The space inside is large enough for seating and worktables. The inner surface is a painted metallic skin that serves as an enveloping whiteboard, which can also hold magnetic pushpins. The large-screen display on the outside can be used for announcements and coordinating activities.

In Fig. 5.2, we show immersion and collaboration within individual and group workspaces for colocation of designers, just outside the Kiva.

5.3 THE PROJECT COMMISSION: HOW DOES A PROJECT BEGIN?

All projects have some type of beginning. Projects are spawned by any of a number of forces, including marketing strategy, management edict, a product sponsor, the awarding of a contract, a system client, inventive ideas, or your own perceived needs to improve an existing product or system.



Fig. 5.1

The Kiva, Virginia Tech's ideation and design studio (courtesy Akshay Sharma, Virginia Tech Department of Industrial Design).



*Fig. 5.2
Individual
and group
designer
workspaces
(courtesy
Akshay
Sharma,
Virginia Tech
Department
of Industrial
Design).*

The commission to undertake a system-oriented project rarely comes from within the UX design or SE group, but instead as a mandate from upper management. The notion of a new product can hatch in the form of a business brief, often from product owner roles (Section 5.4.5) within an established product-oriented organization.

For the project team, things begin with some kind of project commission document or statement that is a proposal or mandate to create the project. Depending on the organizational situation, the project commission document can include boilerplate items, such as project duration, budget, management plan, and personnel roles. A project commission document is likely to be very high-level and ill-defined. It becomes the first responsibility of the UX design team, especially in collaboration with the product owner, to fill in the gaps during the product definition stage and firm up the vision in the system or product concept statement (Section 5.6), which is the project mission statement.

5.4 KEY ROLES

Here we talk about roles, which can be filled by one or more of the cross-functional team people. In a large organization, some people may specialize in a particular role. In a small organization, one person may handle many of these roles.

5.4.1 UX Researcher

A UX researcher performs or leads UX research activities, such as data elicitation, data analysis and modeling, and UX requirement specifications. In many organizations, a UX researcher is also responsible for UX evaluation.

5.4.2 UX Designer

Sometimes called an interaction designer, the UX designer performs or leads UX design, such as design creation, conceptual design, and/or design production, for user interaction. UX designer can also refer to anyone on the UX design team.

5.4.3 Graphic or Visual Designer

A graphic designer is often responsible for visual communication, layout, branding, style, and pixel-perfect visual compositions.

5.4.4 UX Analyst or Evaluator

The UX analyst or evaluator performs UX evaluation. Often on small teams, the same person may perform the duties of UX researcher and UX analyst.

5.4.5 Product Owner or Manager, a Cross-Functional Role

A person in the product manager role (often referred to as the PM) is the so-called owner or chief executive officer of a feature, product, or portfolio of products (just as a person in the UX role is typically responsible for the conceptualization, design, prototyping, and evaluation of a feature, product, or ecology of products). This role embodies the business perspective in the development of a product or system and is responsible for specifying the requirements or capabilities needed by the system to be successful in the market (in the form of a product requirements document in the traditional product development lifecycles and in the form of user stories in an agile approach). UX roles work closely with this role in developing and later validating through UX research these requirements or capabilities.

Typically, the product owner:

- Is close to users
- Knows the product context within the client's domain, business strategy, and competition
- Is responsible for pursuing business goals and for the success of the product
- Maintains the overall vision in the design
- Is in charge of writing/creating:
 - The product concept statement ([Section 5.6](#))
 - User stories, short narratives describing features needed (see [Section 10.2](#))
 - User personas, narrative descriptions of hypothetical users in a target role for design (see [Section 13.3.3.2](#))
- Works with SE in planning implementation sprints and managing prioritized backlog of agile user stories (see [Chapter 26](#))

5.4.6 Software Engineer, a Cross-Functional Role

The role of software engineer is the other key cross-functional role that we have already introduced. This role is responsible for realizing product manager requirements and UX specifications in code.

5.4.7 Project or Program Manager, a Cross-Functional Role

The program manager (often referred to as the Technical Program Manager or TPM) is responsible for coordinating the efforts of all cross-functional roles (product manager, UX, SE, documentation, etc.) and their process lifecycles to converge into a single product release timeline. This person essentially keeps the project on track.

5.5 INTRODUCING A RUNNING EXAMPLE

As a running example to illustrate the ideas in the text, we introduce a hypothetical public ticket sales system for selling tickets for entertainment and other events. This project is a good example of a combined consumer product and an enterprise system. It is an enterprise system for the client, but it is used much like a consumer product by the general public.

5.5.1 The Existing System: Middleburg University Ticket Transaction Service

Middleburg, a small town in middle America, is home to Middleburg University (MU), a state university that operates a service called the Middleburg University Ticket Transaction Service (MUTTS). MUTTS has operated successfully for several years as a central campus ticket office, where people buy tickets for entertainment events, including concerts, plays, and special presentations by public speakers. Through this office, MUTTS makes arrangements with event sponsors and sells tickets to various customers. There is considerable interest in improving and expanding MUTTS. This information about MUTTS is the type you discover in data gathering (see [Chapter 6](#)).

The current business process suffers from numerous drawbacks:

- Until recently, all customers had to go to one location, the MUTTS ticket office, to buy tickets in person.

- MUTTS has now partnered with [Tickets4ever.com](#) as a national online ticket distribution platform. However, [Tickets4ever.com](#) suffers from low reliability and has a reputation of providing a poor user experience.
- Current operation of MUTTS involves multiple systems that do not work well together.
- The ability to rapidly hire temporary ticket sellers to meet periodic high demand is hampered by university and state hiring policies.

The desire to expand the business coincides with a number of other dynamics currently affecting MUTTS and MU:

- The supervisor of MUTTS wishes to expand revenue-generating activities.
- There is a general consensus that the new system should provide a much-improved user experience in several areas.
- To leverage their increasing national academic and athletic prominence, the university is seeking a comprehensive customized solution that includes the integration of tickets for athletic events. Currently, tickets to athletic events are managed by an entirely different department.
- By including tickets for athletic events that generate significant revenue, MUTTS will have access to resources to support other aspects of their expansion.
- The university is undergoing a strategic initiative for unified branding across all its departments and activities. The university administration is receptive to creative design solutions for MUTTS to support this rebranding effort.

5.5.2 The Proposed New System: The Ticket Kiosk System

The working name for the new system is the Ticket Kiosk System, which was commissioned in a business brief by the university administration and MUTTS management. A high-level decision has been made to mandate the notion of a ticket kiosk as the core of this project. As designers, however, we need to keep an open mind and be prepared to steer the project toward a better solution, if one arises, as we get immersed in the problem space.

Early on, the scope of the project must be addressed in the business brief as a road map for all that is to follow. For example, does the design cover kiosk maintenance, payment arrangements, arrangement of agreements for purchase of tickets to outside organizations, and so on? These issues reflect other work¹ roles in the ecology of the system, each of which, if included in the project, will have its own set of tasks to be designed for, especially in the interaction design.

¹The term *work* broadly signifies anything people/users do while using products or systems (see Section 6.2).

5.5.3 Rationale for Kiosks

The administrators of MU and MUTTS wanted to expand its scope and add more ticket office locations, but it is expensive to rent space in business buildings around town, and the kind of small space it needs is rarely available. Therefore, they have decided to switch the business from ticket windows to kiosks, which can be placed in many more locations across campus and around town.

Middleburg has a reliable and well-used public transportation system operated by Middleburg Transit, Inc. There are several bus stops, including at the library and the shopping mall, where buses come and go every few minutes with good-sized crowds getting on and off. Most such locations have space for a kiosk at a reasonable leasing fee.

Management expects an increase in sales due to the increased availability (kiosks in many places) and increased accessibility (open continuously). Also, there may be cost savings because a kiosk requires no personnel at the sales outlets.

5.6 THE PRODUCT/SYSTEM CONCEPT STATEMENT

It is the job of the product owner (Section 5.4.5) and the UX team to crystallize the product/system concept in a clear project mission statement addressed to all stakeholders. For simplicity, we use product concept statement as a general term regardless of whether you are designing a consumer product or an enterprise system.

A product concept statement is like a press release for the product. In 2016, Stephenie Landry, vice president of Amazon Prime Now, began a project by writing the product concept statement as a press release and working backward from there to deduce high-level needs, requirements, and design. In the product concept statement for Amazon Prime Now, she promised a user experience that would be “magical” (Landry, 2016).

Tony Fadell, creator of the Apple iPod (Moore, 2017), says product design and marketing are closely related. He also creates a press release as one of the first steps in designing a new product.

The nature of a product concept statement. A product concept statement is:

- Concise, typically 100 to 150 words in length
- A mission statement for the project team, a vision statement, or a mandate
- A way to explain the product to stakeholders and outsiders
- A yardstick to help set focus and scope for development internally
- Not easy to write (especially a good one)

Because of the need for the statement to be short and concise, every word is important. A concept statement is not just written; it is iterated and refined to make it as clear and specific as possible:

- Aim at a very broad audience, including high-level management, marketing, the board of directors, stockholders, and even the general public.
- Refine and update as necessary as you proceed through the lifecycle activities.
- Post your product concept statement in your design studio as soon as possible. It will guide all UX design activities.

What is in a product concept statement? An effective product concept statement answers at least the following questions:

- What is the product or system name?
- Who are the users?
- How will they use it?
- Enterprise system: What problem(s) will the system solve (broadly including business objectives)?
- Consumer product: What are the major attractions to, or distinguishing features of, this product?
- What is the design vision, and what are the emotional impact goals? In other words, what experience will the product or system provide to the user?

Example: A Product Concept Statement for the Ticket Kiosk System

The Ticket Kiosk System will replace the old ticket retail system, the MUTTS, by providing round-the-clock distributed kiosk service to the general public. This service includes access to comprehensive event information and the capability to rapidly purchase tickets for local events, such as concerts, movies, and the performing arts. The new system includes a significant expansion of scope to include ticket distribution for the entire MU athletic program. Transportation tickets will also be available, along with directions and parking information for specific venues. Compared to conventional ticket outlets, the Ticket Kiosk System will reduce waiting time and offer far more extensive information about events. A focus on innovative design will enhance the MU public profile while fostering the spirit of the MU community and offering the customer a quality interaction experience. (136 words)

Iteration and refinement. Your product concept statement will evolve as you proceed with the project. For example, “far more extensive information about events” can be made more specific by saying “extensive information, including images, movie trailers, and reviews of events.” Also, we did not mention security and privacy, but these are important business concerns. Similarly, the point about “focus on innovative design” can be elaborated by saying “the goal of innovative design is to reinvent the experience of interacting with a kiosk by providing an engaging and enjoyable transaction experience.”

Concept statements for large systems. For very large systems, you may need a system concept statement for each major sub-system, if each one has its own goals and mission statement.

5.6.1 Introduction to Process-Related Exercises

After each UX design activity is discussed in the process chapters, we often give you an exercise to try your hand at doing that activity. These exercises are intended as in-class activities, if your learning environment permits. These are different from team project assignments that we suggest, which are assignments outside of class, but they should use the same teams. These team projects are semester-long, out-of-class projects envisioned to involve the complete design of a midsize interactive system with real clients. In contrast, the in-class exercises are meant to focus on learning the techniques or methods rather than on completeness.

5.6.1.1 How to approach these exercises

The exercises are for learning, not for producing a product, so you will not have to complete every detail if you think you have mastered the concept. You should be able to learn most of what you need from an exercise in an hour or so. In the case of a team within a classroom setting, this means that you can do the exercises as in-class activities, breaking out into teams and working in parallel. If necessary, you can finish the exercise as homework before the next class. This has the advantages of working next to other teams with similar goals and problems and of having an instructor present who can move among teams as a consultant and mentor. We recommend that student team deliverables be prepared in summary form for presentation to the rest of the class, if you have time, so that each team can learn from the others.

5.6.1.2 Choosing a product or system for these exercises

If you are a student in a UX class, the exercises in the book chapters will be done as in-class exercises, and the instructor will suggest a product or system to use. Readers who are not part of a class should choose their own product or

system. This should take some thought because you will use the same choice for most of the exercises throughout the process chapters.

Your choice of a target application system for the exercises should be gauged toward the goal of learning, which means choosing something that is the right size. Avoid applications that are too large or complex; choose something for which the semantics and functionality are relatively easy to understand. Avoid systems that are too small or too simple because they may not support the process activities very well.

You will need to identify at least a few user roles (see [Section 8.2](#)) and five or six somewhat different types of user tasks for each of these roles. Look for something with an interesting ecology. For example, an ecommerce website for ordering merchandise will have users from the public placing the orders and employee user roles processing the orders. We used these criteria in choosing the Ticket Kiosk System as the book example.

For practitioner teams in a real development organization, we recommend against using a real development project for these exercises. There is no sense in introducing the pressure to produce a real design and the risk of failure into this learning process.

Exercise 5.1 Product Concept Statement for a Product or System of Your Choice

Your first exercise is here. Active learning means learning by doing. The best way to learn the processes described in this book is by doing the exercises. We have organized your participation in the process chapters at three levels: examples for you to follow in the text, a more or less parallel set of exercises to do with a group or on your own, and a set of extensive and specific team project assignments (<http://www.theuxbook.com/>). We start you off here with an easy one.

Goal: Get practice in writing a concise product concept statement.

Activities: Write a product concept statement for a product or system of your choice. Iterate and polish it. The 150 or fewer words you write here will be among the most important words in the whole project, so write thoughtfully.

Deliverables: Submit your product concept statement.

Schedule: Given the simplicity of the domain, we expect you can get what you need from this exercise in about 30 minutes.

5.7 SEGUE: WELCOME TO THE PROCESS CHAPTERS

We now move into [Part 2](#) of the book, where we do UX research to understand needs and requirements. [Part 3](#) is about design, followed by [Part 4](#), which is about prototyping. The process chapters are completed in [Part 5](#), about UX evaluation.